

5G/6G Communication Systems- ECA5601(LAB)

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Experiment 12: NB-IoT vs LTE-M Throughput Simulation

Aim: To simulate and compare the performance of NB-IoT and LTE-M communication technologies in terms of Throughput (Mbps), Latency (ms), and Packet Delivery Ratio (%) using MATLAB.

Objective 1:

Program:

```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Simulated Throughput (Mbps)
NB_IoT = 0.05*(1-exp(-SNR/8));
LTE_M = 1.2*(1-exp(-SNR/8));

disp('SNR (dB) NB-IoT(Mbps) LTE-M(Mbps)')
disp([SNR' NB_IoT' LTE_M'])
figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(SNR,NB_IoT,'-o','LineWidth',2)
title('NB-IoT Throughput vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)')
```

grid on

% ----- Graph 2 -----

subplot(2,1,2)

plot(SNR,LTE_M,'-s','LineWidth',2)

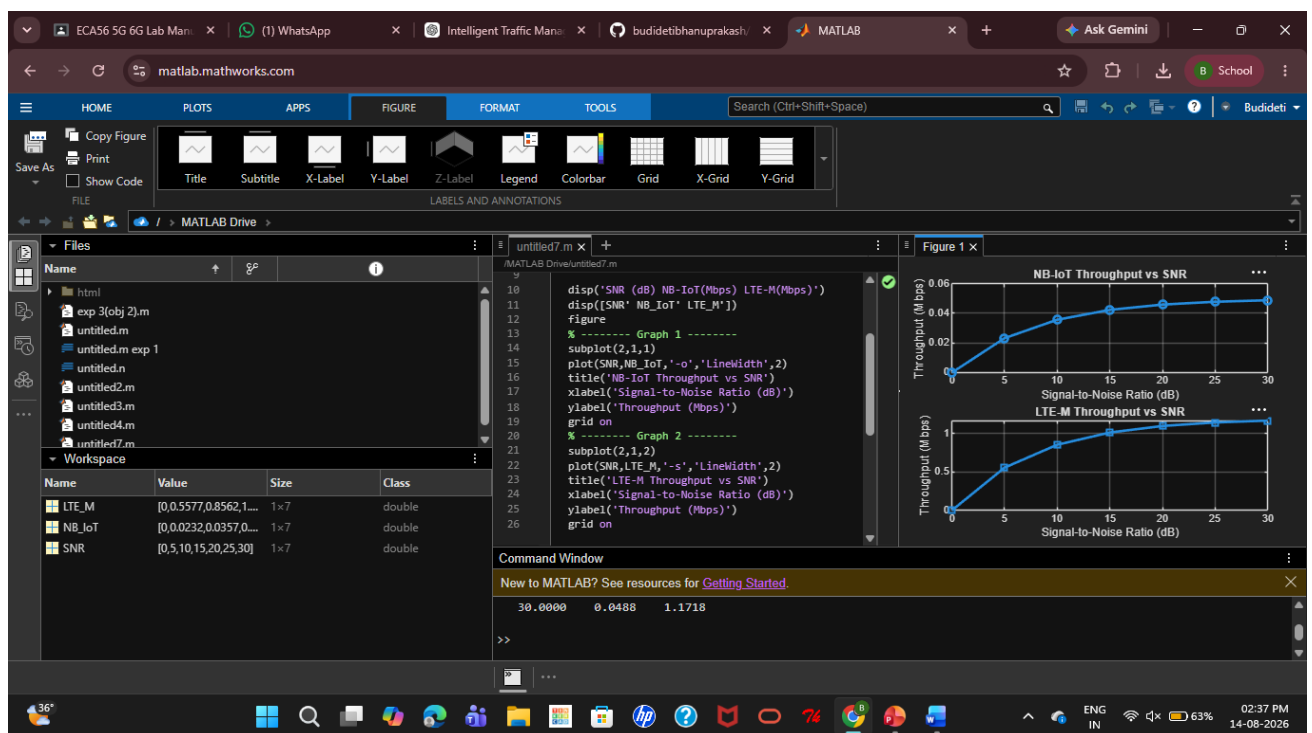
title('LTE-M Throughput vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Throughput (Mbps)')

grid on

Output:



Objective 2:

Program:

clc;

clear;

close all;

% SNR values (dB)

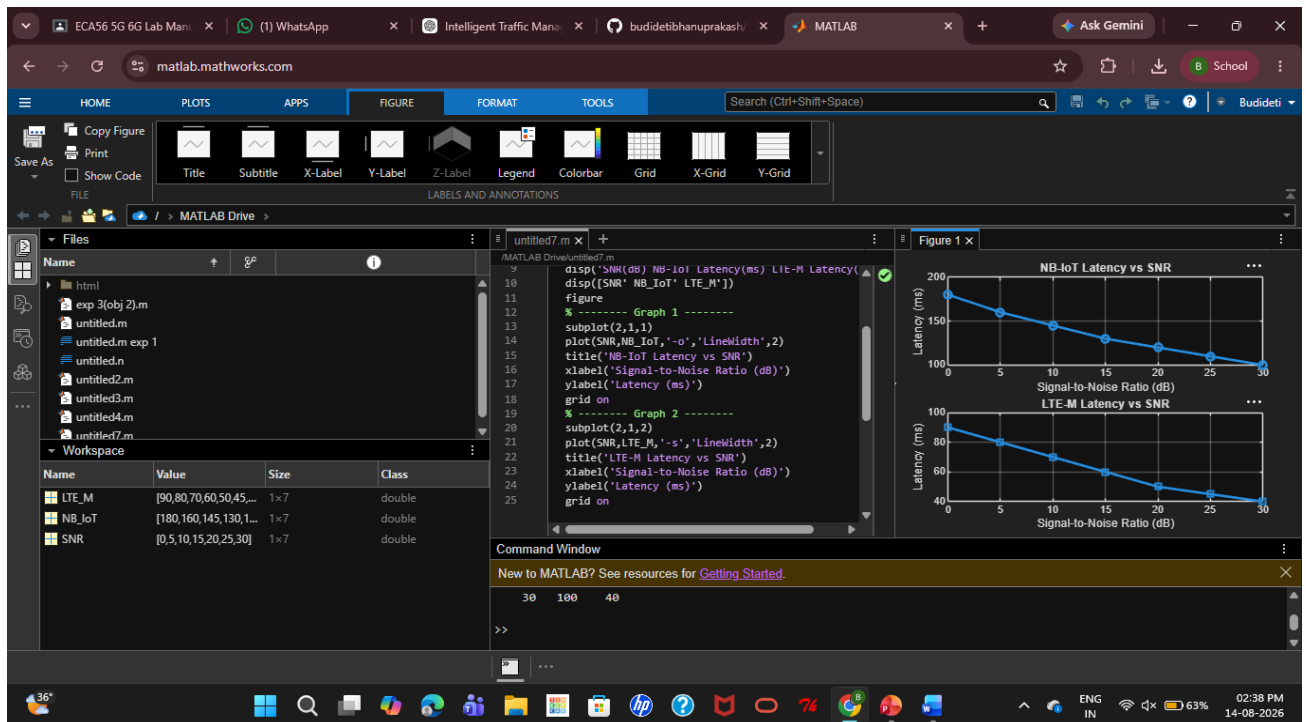
SNR = 0:5:30;

```

% Simulated Latency (ms)
NB_IoT = [180 160 145 130 120 110 100];
LTE_M = [90 80 70 60 50 45 40];
disp('SNR(dB) NB-IoT Latency(ms) LTE-M Latency(ms)')
disp([SNR' NB_IoT' LTE_M'])
figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(SNR,NB_IoT,'-o','LineWidth',2)
title('NB-IoT Latency vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Latency (ms)')
grid on
% ----- Graph 2 -----
subplot(2,1,2)
plot(SNR,LTE_M,'-s','LineWidth',2)
title('LTE-M Latency vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Latency (ms)')
grid on

```

Output:



Objective 3:

MATLAB Code:

```
clc;

clear;

close all;

% SNR values (dB)
SNR = 0:5:30;

% Simulated Packet Delivery Ratio (%)
NB_IoT = [82 86 89 92 95 97 99];
LTE_M = [88 91 94 96 98 99 100];

disp('SNR(dB) NB-IoT PDR(%) LTE-M PDR(%))')
disp([SNR' NB_IoT' LTE_M'])

figure

% ----- Graph 1 -----
subplot(2,1,1)
plot(SNR,NB_IoT,'-o','LineWidth',2)
title('NB-IoT Packet Delivery Ratio vs SNR')
```

```
xlabel('Signal-to-Noise Ratio (dB)')
```

```
ylabel('Packet Delivery Ratio (%)')
```

```
grid on
```

```
% ----- Graph 2 -----
```

```
subplot(2,1,2)
```

```
plot(SNR,LTE_M,'-s','LineWidth',2)
```

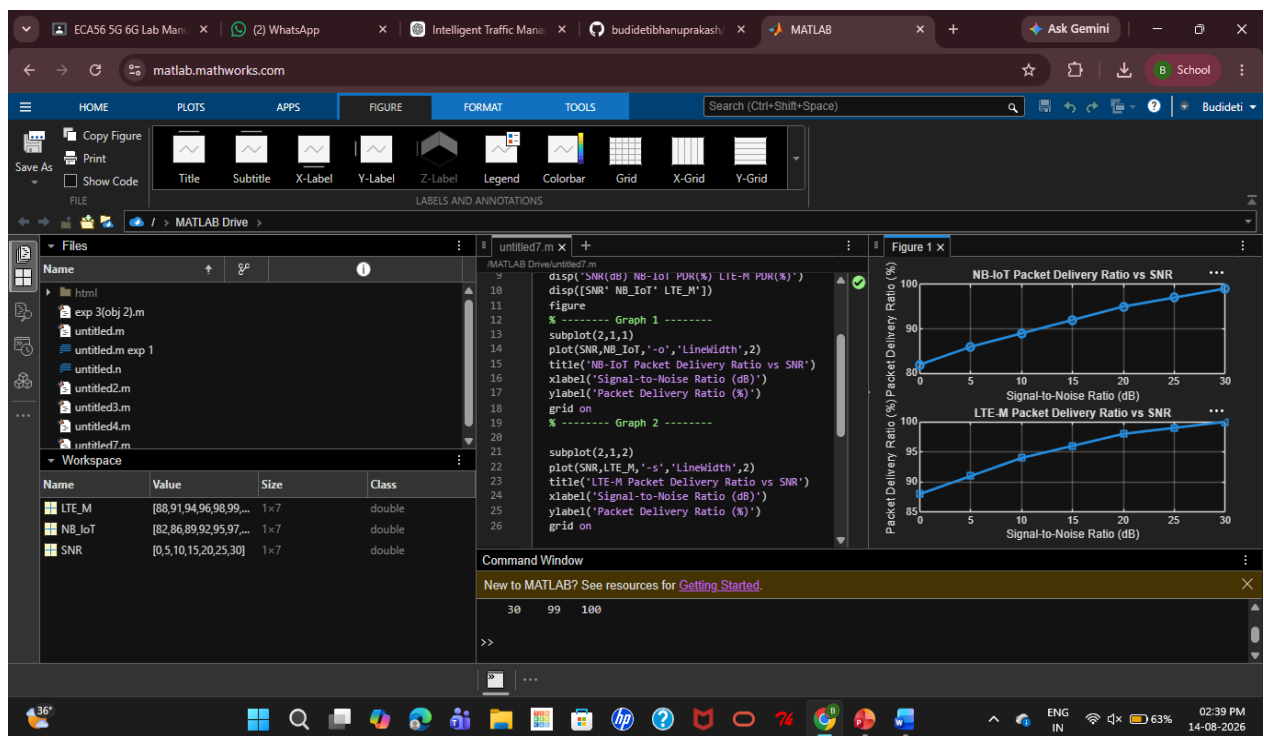
```
title('LTE-M Packet Delivery Ratio vs SNR')
```

```
xlabel('Signal-to-Noise Ratio (dB)')
```

```
ylabel('Packet Delivery Ratio (%)')
```

```
grid on
```

Output:



Result:

The NB-IoT and LTE-M systems were successfully simulated in MATLAB. Their Throughput, Latency, and Packet Delivery Ratio were compared under varying network conditions, and the performance of both technologies was evaluated.